

The scalar algebra contradicts the Capelli equivalence

Disproof of Conjecture 00000002755

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Convention and claim. Use the standard Capelli index counting the alternating variables:

$$C_n(x_1, \dots, x_n; y_1, \dots, y_{n-1}) = \sum_{\sigma \in S_n} \text{sgn}(\sigma) x_{\sigma(1)} y_1 x_{\sigma(2)} \cdots y_{n-1} x_{\sigma(n)}.$$

In particular $C_1(x_1) = x_1$. The conjecture asserts that an algebra satisfies $C_n = 0$ if and only if its PI exponent is at most n . We disprove this equivalence at $n = 1$ for the unital characteristic-zero algebra $A = \mathbb{Q}$. No restriction excluding this algebra or this index occurs in either version of the problem.

Actual multilinear codimensions. For $m \geq 1$, let P_m be the vector space with basis the noncommutative multilinear words $x_{\sigma(1)} \cdots x_{\sigma(m)}$, $\sigma \in S_m$. Its evaluation map is

$$E_m : P_m \longrightarrow \{\mathbb{Q}^m \rightarrow \mathbb{Q}\}, \quad E_m\left(\sum_{\sigma} a_{\sigma} x_{\sigma(1)} \cdots x_{\sigma(m)}\right)(t) = \sum_{\sigma} a_{\sigma} \prod_{j=1}^m t_{\sigma(j)}.$$

Its kernel is precisely $P_m \cap \text{Id}(A)$, since belonging to the kernel means vanishing on every substitution in the actual algebra. Commutativity gives

$$E_m(p)(t) = \left(\sum_{\sigma} a_{\sigma}\right) \prod_{j=1}^m t_j.$$

Thus its image is the span of $g_m(t) = \prod_j t_j$. This function is nonzero because $g_m(1, \dots, 1) = 1$, and it belongs to the image by evaluating the identity-permutation word. The first isomorphism theorem therefore yields

$$c_m(A) = \dim_{\mathbb{Q}}(P_m / (P_m \cap \text{Id}(A))) = \dim_{\mathbb{Q}} \text{im } E_m = 1.$$

Consequently the genuine PI exponent is $\exp(A) = \lim_{m \rightarrow \infty} \sqrt[m]{c_m(A)} = 1$.

Failure of the equivalence. The proposed bound $\exp(A) \leq 1$ holds. However, $C_1(1) = 1 \neq 0$, so A does not satisfy the Capelli identity of index one. This disproves the stated equivalence and hence the conjecture. For orientation, $C_2 = x_1 y_1 x_2 - x_2 y_1 x_1$ does vanish in this commutative algebra, so its actual minimal index is two. This agrees with the separate “ceiling plus one” clause, which does not repair the false equivalence.

Formal certificate. The Lean project constructs permutation words, their linear evaluation map into all scalar-valued functions, its exact one-dimensional image, and the quotient-by-kernel linear equivalence. It derives all codimensions and the exponent limit, then evaluates the actual index-one polynomial at one. Final theorem axiom audits use only the standard axioms of Lean and Mathlib.